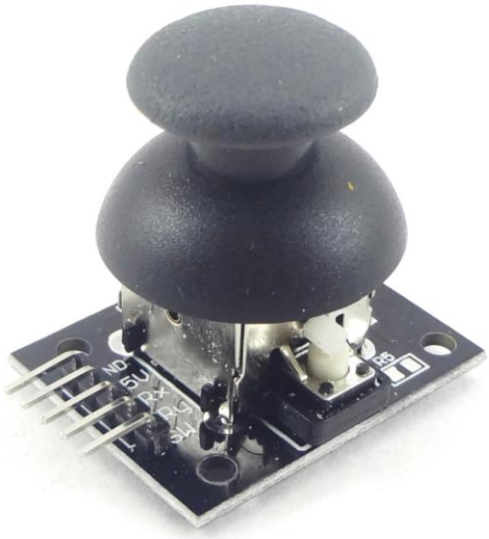
Building the platform

Exercise 1:

- Start building the platform board as a group, you'll need
  - Wooden board
  - Arduino mega
  - Breadboard for electrical circuits
  - LCD (wiring help in file: LCD wiring)
  - Joystick (wiring help in file: Joystick wiring)



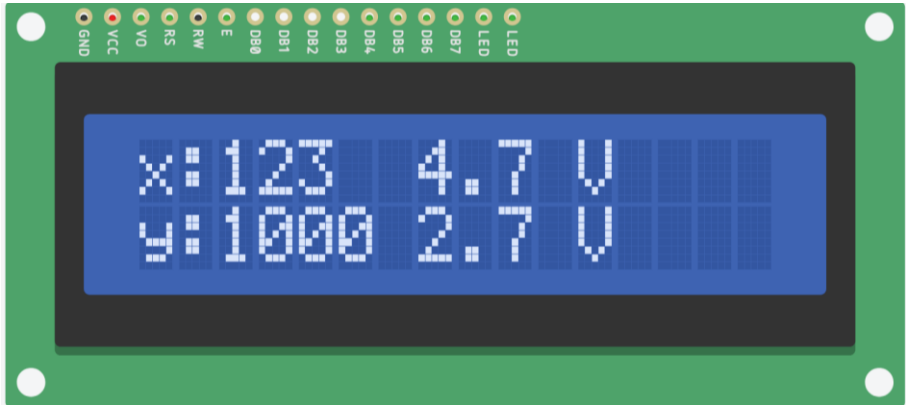
Revision LCD

Exercise 2:

- Create a program that prints into LCD-display
  - the analog value (ie result of AD-conversion) and voltage of joystick y-value
  - the analog value (ie result of AD-conversion) and voltage of joystick x-value

Some recap videos:

- Arduino in general click
- Arduino analog inputs click
- Using LCD with Arduino click



Revision ISR (Interrupt Service Routine)

Exercise 3:

- Continue with your code from last exercise. Add a functionality, where using the ISR and joystick's push button changes between showing:
  - the analog data from exercise 2
  - total amount of button presses



Recap:

- Arduino Interrupts video click
- Attach interrupt reference click

Building the car

Exercise 4:

- Build your car, you'll need
  - 2 motors with cables
  - 2 wheels (not 4!!!)

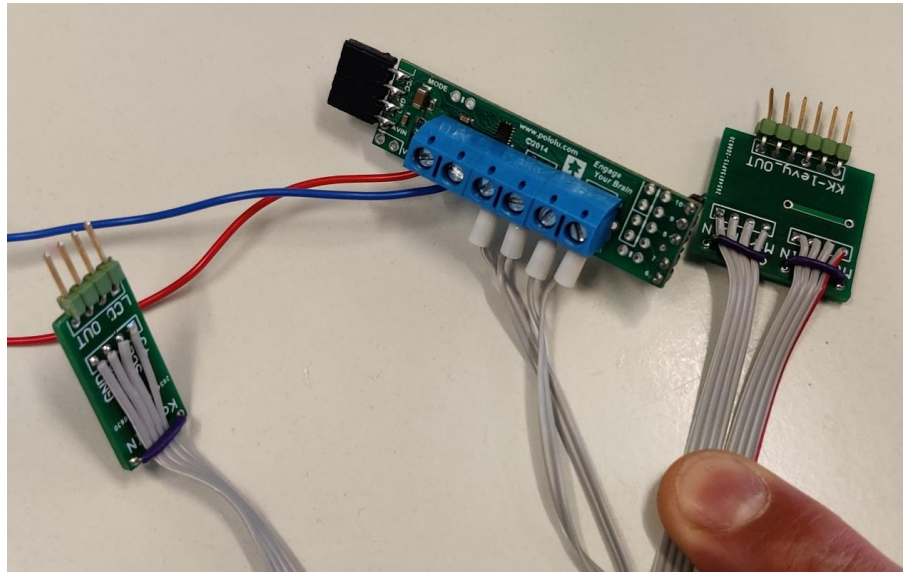
- Make it sturdy but not too big (we'll add a few components on top of the car later on)

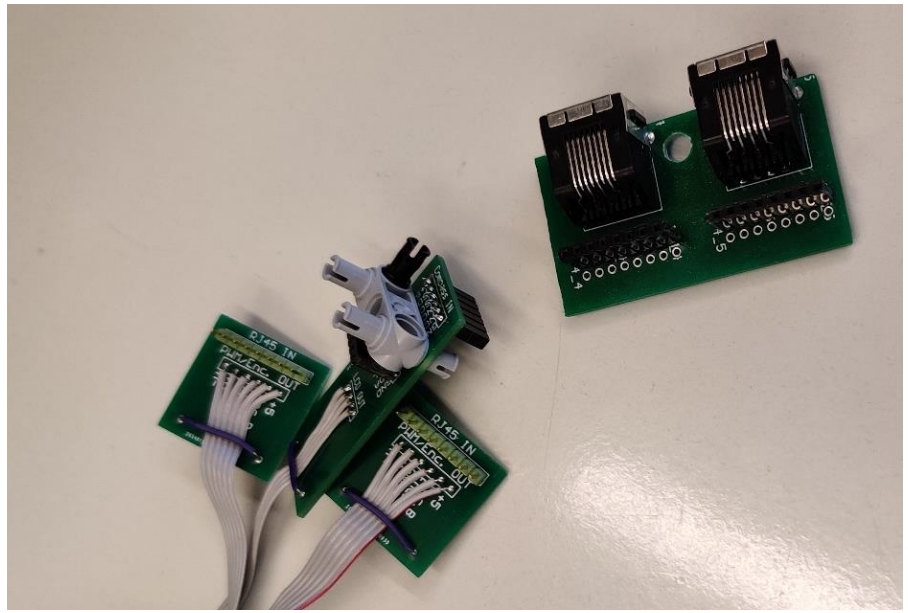


Testing the car

Exercise 5:

- Choose one cable set and test that it is working correctly using a multimeter (ie. connection is not lost from one end of the cable to another). Help?
- Connect your car to the Arduino using the cable set and a motor driver (more info about cables in file: Cable help)
- Download and run motor test code from Moodle
  - Car should be moving forwards and backwards

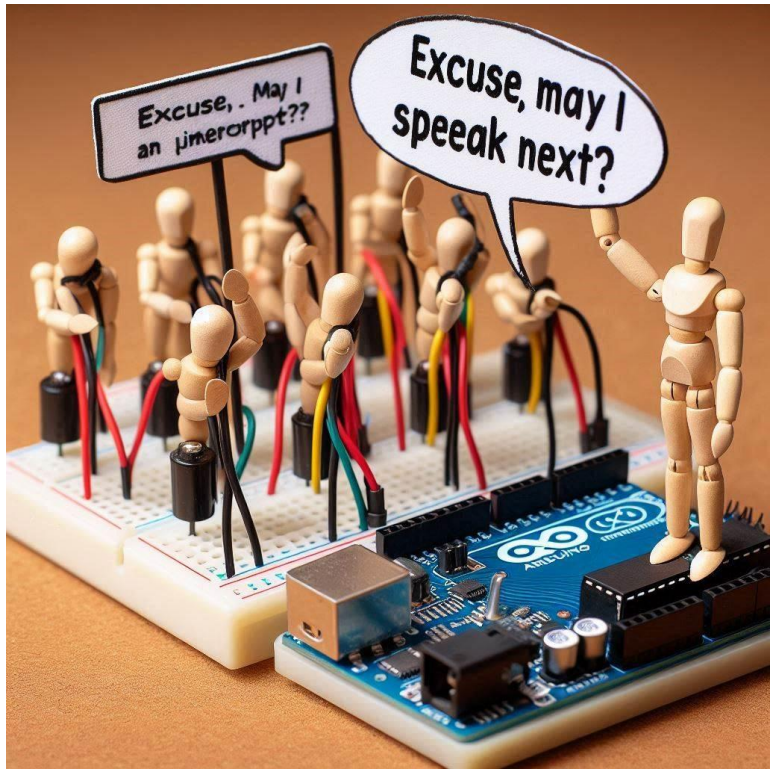




Using the car

Exercise 6:

- Use your code from exercise 3 and combine with it the motor test code according to these instructions:
  - analog value display works the same way as before
  - when changing to the counter display, your Arduino will execute the motor test run (which is in its own function)



Recap:

- Arduino functions click